REPTILE REPRODUCTIVE DISEASES

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1. ANATOMY OF THE REPRODUCTIVE SYSTEM

The female reproductive tract is composed of the paired ovary and oviducts.

The male reproductive tract is composed of paired testicles, epididymis, and ductus deferens and depending on the species, two hemipenes or a single phallus.

See your anatomy and physiology lecture for further details.

2. REPRODUCTIVE DISORDERS IN THE FEMALE

2.1 Follicular statis

Ovarian(preovulatory) follicular stasis in reptiles occurs when mature follicles develop but do not ovulate or resorb, becoming static and eventually degenerative. This condition occurs commonly in lizards, chelonians. It is often a seasonal problem because hormonal cycles but can occur at any time of the year. In captive short-tail pythons has been link ot the absence of male. Metabolic disease such as NSHP may contribute. Follicular stasis might resemble normal folliculogenesis early on. A clear understanding of the reproductive biology of the species, a thorough history, physical examination, and, in some cases, supplemental diagnostic tests may be necessary to determine the need for intervention. Reptiles with follicular stasis may be restless, pacing and climbing, looking for possible mates or nesting sites. Appetite and water intake may be reduced or absent. Coelomic distention with reduced body condition, anorexia, and behavior changes may be noted.

On physical examination careful palpation may confirm preovulatory follicles in squamates. Ovarian follicles tend to be more dorsal and spherical and not as mobile as oviductal eggs, which are usually more oblong and ventral/caudal in the coelom. Plasma biochemical changes often include elevations of total calcium; however, ionized calcium is often normal. Hematology may show a stress leukogram or, in cases of inflammation or infection, a leukocytosis, or leukopenia with left shift due to sequestration may be present. In more chronic cases anemia may be observed. Radiographically squamate follicular stasis often appears as a cluster of spherical, dorsal, homogenous, soft tissue structures that lack a shell. Ultrasonography is preferred for identification of follicular stasis and to differentiate between follicular stasis and poorly shelled postovulatory eggs. Sonographically, vitellogenic follicles show no hyperechoic shell and are uniformly hypoechoic and clustered. Ultrasonography can be utilized to monitor follicular progress. With active resorption or degenerate changes the ova progress from homogenous and hypoechoic to a more mixed (hypoechoic and anechoic) echogenicity. This progresses to a loss of shape, and follicles become less spherical and poorly demarcated. With resorption they continue to reduce in size, but with stasis and degeneration the follicles coalesce with surrounding anechoic fluid often evident. If echogenic free-floating debris is noted within this anechoic fluid the ova are likely degenerate and inflamed, and surgical intervention is indicated. CT and MRI may also be indicated in some cases, and this technology is becoming more accessible. Specifically in managing follicular stasis, CT (versus radiography) allows each ovarian follicle to be individually evaluated. CT is more useful for identifying advanced changes, including distinct horizontal leveling of the contents of degenerating follicles indicating the need for intervention.

Treatment for follicular stasis may differ. If the reptile is a pet or display animal then ovariectomy is preferred. Alternatively, where the patient is considered healthy, a nest box can be provided and the reptile (except snakes) sent home on calcium glubionate (23 mg/ mL) at 1 mL/kg PO SID-BID for 2 to 3 weeks depending on dietary history. Appetite and behavior should be monitored, and reevaluation is recommended in 2 to 3 weeks to determine whether resorption or ovulation has occurred. If the patient is not considered healthy they must be stabilized by providing appropriate thermal support for the species and fluid therapy if indicated. Additional medical therapy may be initiated, including calcium (lizards and chelonians), pain management and antimicrobials if needed before performing ovariectomy

2.2 Dystocia

Dystocia is the inability to successfully expel term eggs or fetuses from the lower reproductive tract. The reasons for dystocia are poorly understood in reptiles but may include abnormal eggs/fetuses, failure of the distal reproductive tract to contract, lack of suitable nesting or birthing sites, and lack of viable embryos. Dystocia is usually a seasonal problem because hormonal cycles, but can occur at any time of the year. Dystocias can be obstructive or nonobstructive. Obstructive dystocia may be caused by a fetal or maternal abnormality. Fetal abnormalities include oversize, malformations, and adhesions. Maternal abnormalities include misshapen pelvis, oviductal stricture, or coelomic pathology, including cystic or cloacal calculi, obstipation, organomegaly, abscesses, or neoplasia. In addition, obstructive dystocias may result from complications during oviposition, such as malpositioning of eggs or broken/damaged eggs. Nonobstructive dystocias can be caused by improper nesting sites or temperature, malnutrition, and dehydration. One of the most common causes of nonobstructive dystocia in lizards and chelonians relates to calcium deficiency. Another cause of dystocia may relate to poor physical condition of the female, specially snakes.

Reptiles with dystocia may be restless, pacing and climbing, looking for possible nesting sites. Oviparous lizards and chelonians may dig in planters or substrate. Female aquatic turtles may spend extended time out of the water. Appetite and water intake may be reduced or absent. Coelomic distention with reduced body condition and anorexia may be noted. Tenesmus might be present. Eggs in squamates can often be differentiated from follicles because they are firm, larger, more oblong in shape, and more caudally located in the coelom. Chelonians are more difficult to palpate, but in smaller species digital palpation in the prefemoral fossae might be possible.

Plasma biochemical changes often include elevations of total calcium; however, ionized calcium is often normal. Complete blood count may show a stress leukocytosis or, in cases of inflammatory/infectious change, a leukocytosis with left shift. In more chronic cases anemia and leukopenia may be observed. Radiographs are important to exclude obstruction (before the use of oxytocin) and confirm egg numbers, shell abnormalities (e.g., thickening, broken, size) and ectopic eggs. With viviparous squamates radiography can be useful late in gestation when skeletal structures become visible. Typically, viable fetuses are coiled, whereas an uncoiled or linear appearance likely indicates fetal death. Radiography may also help to identify abnormalities associated with nutritional or renal secondary hyperparathyroidism. Ultrasonography (US) can differentiate between follicular stasis and postovulatory eggs: postovulatory eggs exhibit hyperechoic mineralized outer shells, and, in squamates, a more oblong appearance. A small amount of anechoic fluid may be visualized around and between eggs. In viviparous squamates US can identify fetal movement and heartbeat (see Fig. CT allows individual egg evaluation and can demonstrate indicators of dystocia described above with radiography and US.

Dystocia is rarely, the acute emergency that it is in mammals and birds and the treatment will depend on the physical condition of the patient and the severity of the clinical signs. Initial treatment should be aimed at stabilizing the patient. Supportive care including fluid therapy and thermal support might be needed. For non-obstructive dystocia, if the female appears healthy, a nest box and other environmental needs can be provided, while monitoring closely and reevaluating within a few days to 1 to 2 weeks unless tenesmus or deterioration becomes evident. If these fails, and there is still no evidence of obstruction, medical management treatment with thermal support, fluids, calcium and/or oxytocin might be required. Chelonians are more responsive to oxytocin, even in more chronic cases of dystocia. Oxytocin dosing differs considerably with chelonians responding well to lower doses (1–4 IU/kg) compared with higher doses necessary for squamates (10–20 IU/kg) SQ, IM or IV slow. Dosing can be repeated and the dose increased. Prostaglandin F2 alpha (1.5 mg/ kg) administered SQ was used in combination with oxytocin (7.5 IU/ kg) and found to be effective in inducing oviposition in red-eared sliders. Due to the linear anatomy of snakes, egg manipulation may be possible and is noninvasive. If the first egg moves freely toward the vent, a vaginal speculum can be used to dilate the cloaca and allow visualization of the egg at the oviduct opening. Sometimes it can be gently grasped and removed with forceps, or ovo-centesis, using a large-gauge needle (14–18 G), may be necessary to reduce size. The collapsed egg can then be removed with forceps or hemostats, and the procedure repeated for the remaining eggs. This procedure is only recommended in lizards and chelonians if the egg is visible within the cloaca. In these cases, however, the egg often breaks, requiring piece-meal removal of shell fragments. Cloacoscopy may be utilized to manipulate retained eggs/fetuses out of the cloaca if medical and/or less-invasive techniques have failed or are not appropriate. Percutaneous ovocentesis is a technique that can be considered in snakes. This technique has risks like possible leakage of egg contents into the coelom or iatrogenic needle trauma. Deep sedation to general anesthesia is recommended. The snake will usually pass the egg within 12 to 24 hours of aspiration. Eggs cranial to the collapsed egg may pass on their own after the first egg passes or may also need aspiration. Postoperative analgesics and antimicrobials might be needed. Eggs retained more than several weeks are not likely to aspirate as the contents solidify, and surgical removal is required. Surgical management by ovariosalpingectomy offers a permanent solution for resolving dystocia in chelonians and lizards. Salpingotomy can be performed in breeding animals if less-invasive methods have failed.

2..3 Yolk peritonitis

Egg yolk peritonitis occurs when yolk material is accidentally deposited within the peritoneal cavity. Many factors can lead to this condition. Salpingitis can lead to reverse peristalsis and spilling of the oviduct content through the funnel of the infundibulum and into the intestinal peritoneal cavity. Rupture of the oviduct can also result in leakage of egg material into the peritoneal cavity. Trauma can lead to ovarian follicle rupture and leakage of yolk into the coelom. Egg yolk peritonitis can be aseptic, but in most cases bacteria quickly colonize the rich yolk resulting in septic peritonitis.

Clinical signs include depression, lethargy, weight loss, abdominal distention, and dyspnea. The diagnosis is based on history, physical exam findings, clinical signs, and the fluid evaluation (including cytology) of the coelomic exudate. Radiographs may be useful, but the presence of ascites often obscures serosal detail. In these cases, ultrasonography is best for assessing the coelomic cavity and the reproductive tract.

Supportive care with fluid therapy, thermal and nutritional support might be needed. Antibiotics might be indicated to treat underlying infection or prevent it in severe cases.. Refractory cases or cases in which a large amount of inspissated yolk is present within the peritoneal cavity may need coeliotomy to lavage the intestinal peritoneal cavity.

2.4 Oophoritis and salpingitis

Oophoritis or ovarian inflammation is often caused by bacterial agents, and less commonly by viral, and rarely fungal or parasitic agents. Oophoritis usually results from spreading infections often results from hematogenous spread. Affected ovaries are usually enlarged, wrinkled, discolored, and hemorrhagic.

The clinical include depression, anorexia, weight loss, abdominal distention, dyspnea, and sudden death. The diagnosis of oophoritis can be challenging. Ultrasound and or CT, followed by endoscopic examination are the most reliable diagnostic tools. Exploratory celiotomy and ovariectomy can be both diagnostic and therapeutic, for a non-breeding reptile. Antibiotics, anti-inflammatories and supportive care (fluids, nutritional support, and warmth) should be administered as indicated. GnRH agonists has been shown to be poorly effective in reptiles based on the few studies performed.

Salpingitis or inflammation of the oviducts can involve the whole organ, or in some cases it may be localized to the uterus (metritis). Viral, bacterial, fungal, and parasitic infections can affect the oviduct. The organisms can originate at the cloaca, peritoneal cavity, or blood. Noninfectious causes include trauma and inspissated yolk. Affected oviducts are usually enlarged, reddened, and friable. Salpingitis are common causes of dystocia or can result as sequela to dystocia.

Clinical signs include depression, weight loss, abdominal distention, tenesmus, infertility, and abnormally shaped and/or colored eggs. Blood work could show evidence of inflammation, anemia of chronic disease or other non-specific findings. Radiography, and ultrasonography and CT can be useful diagnostic tools. However, early diagnosis can be challenging. Multiple organ systems are often affected at advance stage due to oviductal rupture and/or septicemia. Medical treatment is based in antibiotics, anti-inflammatories and supportive care. In severe cases ovariosalpingectomy may be required.

2.5 Ovarian and oviductal neoplasia

Reproductive neoplasms are most commonly found in lizards, except for granulosa cell tumors, which are more common in snakes. Reproductive neoplasia is exceptionally rare in chelonians or crocodilians. Ovarian adenocarcinomas are the most common reproductive tumors found and have a higher incidence of metastasis in snakes compared with lizards. Many individual case reports and reviews have reported ovarian carcinomas, fibromas, hemangiomas, dysgerminomas (red-eared sliders), and teratomas. Oviductal neoplasms reported include leiomyosarcomas and leiomyomas. The clinical signs are highly variable, depending on the type of neoplasia and the stage. In most cases endoscopy or coeliotomy with biopsy will be necessary to confirm an antemortem diagnosis. Ovariosalpingohysterectomy may be curative if diagnosed early, especially for benign disease. In cases where surgical removal of the affected reproductive tract is not possible, or metastasis has occurred, oncologic therapy will be aimed at prolonging the patient’s quality of life rather than curing the patient.

2.6 Oviductal prolapse

Prolapse of oviductal tissues can occur as a complication to chronic egg laying, oviductal inertia, egg binding and dystocia, and space-occupying masses. Prolapses often occur secondary to excessive or prolonged straining from undesired behaviors such as self-stimulation and from medical causes. Treatment is aimed at stabilizing the patient with fluid therapy, thermal and nutritional support, and pain management. The prolapsed tissues must be protected and cleansed with warm saline and/or hypertonic solutions. If the prolapsed tissues are viable, should be gently replaced as soon as possible. If the tissues cannot be immediately replaced, they should be kept lubricated such as with a sterile surgical lubricant gel until surgery. Sutures across the vent may be necessary to keep the tissues in place after reduction, while allowing enough space between the sutures to defecate. Bloodwork and survey radiographs are recommended to evaluate underlying causes. Antibiotics and anti-inflammatories should be administered.

3 REPRODUCTIVE DISORDERS IN THE MALE

3.1 Orchitis

Inflammation of the testicles (orchitis) can be due to non-infectious and infectious diseases. Bacterial infections can result from extension of infections from the air sacs or peritoneal cavity, or hematogenous spread. Ascending infections from the phallus or cloaca can also result in primary orchitis. Affected testicles are usually enlarged, reddened, and may contain multifocal yellow-white foci.

The clinical signs are usually nonspecific and may only show clinical signs if the disease is systemic. In breeding reptiles the most common clinical finding is decreased fertility. The diagnosis of orchitis is based on endoscopic examination and testicular biopsy collection for histopathologic evaluation and culture and susceptibility testing. Antibiotics and anti-inflammatories might be required.

3.2 Neoplasia

Testicular neoplasia is a rare finding in reptiles, and seminomas are the most common neoplasia found. There are several testicular tumors that have been reported. These include interstitial cell adenomas, Sertoli and Leydig cell tumors.

3.3 Phallus or hemipene prolapse

Typically prolapse occurs as a result of trauma. Traumatized tissue becomes engorged, making retraction difficult, and the exposed tissue is subjected to further damage, infection, and necrosis. The prolapsed organ should be thoroughly cleaned and disinfected. Systemic and/or local antibiotic therapy should be instituted if suspected bacterial infectionn. The organ can be replaced in the cloaca (phallus) or hemipenal sac. If severe trauma and/or necrosis are present amputation may be the best treatment option.

REFERENCES [1-5]

1. Di Girolamo N, Selleri P. Reproductive Disorders in Snakes. *Vet Clin North Am Exot Anim Pract* 2017;20:391-409.
2. Gumpenberger M. Diagnostic Imaging of Reproductive Tract Disorders in Reptiles. *Vet Clin North Am Exot Anim Pract* 2017;20:327-343.
3. Innis CJ, DeNardo DF. Theriogenology In: Divers SJ,Stahl SJ, eds. *Maders Reptile and Amphibian Medicine and Surgery,*. 3rd ed ed. St. Louis, MO: Elsevier, 2019;849-893.
4. Knotek Z, Cermakova E, Oliveri M. Reproductive Medicine in Lizards. *Vet Clin North Am Exot Anim Pract* 2017;20:411-438.
5. Perry SM, Mitchell MA. Reproductive Medicine in Freshwater Turtles and Land Tortoises. *Vet Clin North Am Exot Anim Pract* 2017;20:371-389.

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1. ANATOMY OF THE URINARY SYSTEM

The urinary system is composed of paired kidneys, ureters that end in the cloaca, and in some species a bladder and accessory bladders that connects to the cloaca. See your anatomy and physiology lecture for further details.

2. ETIOLOGY AND PATHOPHYSIOLOGY OF URINARY DISORDERS

Renal disease in birds is an under-recognized problem. There are a number of different etiologies associated:

Infectious

Bacterial infections are one of the most common causes of nephritis in reptiles and can occur from a systemic infection. It can also occur from ascending infection since the ureters empty in the cloaca. Gram-negative organism are most common*. Salmonella* spp*., Pseudomonas* spp*., Aeromonas* spp*.,* and *Mycobacterium* spp*.,* and *Leptospira* spp. have been reported.

Viral infections are usually multisystemic, although some viruses like demonstrate a trophism for the kidney like picornavirus. Other viruses that have been associated with renal lesions include ranavirus, arenavirus and adenovirus.

Fungal infections affecting the kidneys are rare and most commonly associated with an systemic infection. *Geotrichum candidum* , *Penicillium griseofulvum, and Chamaeleomyces granulomatis* have been reported. Microsporidia, most commonly including *Encephalitozoon* spp*., Pleistophora* spp*., and Nosema* spp*.*, has also been reported in a number of reptile species with variable effects in the kidney.

Parasitic nephritis due to flagellated protozoa are common in reptiles. These infect the intestinal lumen and can ascend into the urinary tract. *Spironucleus* (synonymous with Hexamita) spp. are most common and have a direct life cycle, infecting the host through ingestion of contaminated feces or urine. *Spironucleus* spp. can infect the renal tubules causing glomerular lesions and generalized tubulointerstitial nephritis with heterophilic inflammation. Intranuclear coccidia have been sporadically reported to cause systemic disease in chelonians; however, organisms are commonly found in the urinary tract resulting in inflammation of the kidneys and urinary bladder with opportunistic secondary bacterial infection. Other protozoa*, such as Entamoeba invadens, Cryptosporidium* spp*.,* and *Eimeria* spp., can infect the kidney, usually as an extension of gastrointestinal disease. *Klossiella* spp. affects often the ducts and ureters and rarely cause pathology or clinical disease. Myxosporidia (*Myxidium* spp.) trematodes (*Styphlodora renalis* and *S. horrida*) and spirorchid flukes have also been found to cause renal pathology.

Inflammatory

Renal amyloidosis is exceptionally rare in reptiles and there are only a few reports.

Toxicity

Renal disease may occur as a secondary condition due to toxicity. Drug-induced renal disease is primarily related to the use of antibiotic and NSAIDs. Among antibiotic agents, aminoglycosides like gentamicin and amikacin may be the most problematic. NSAIDs, mostly meloxicam, have been less often associated with renal toxicity but overdose can result in renal failure.

Anomalous

There are some described congenital abnormalities in reptiles, mostly involving agenesis of parts of the kidney in snakes. Kidney cysts are exceptionally rare.

Metabolic

Dehydration is an important contributor to renal disease. Severe or persistent dehydration increases resorption of water causing a subsequent reduction in urine flow. As uric acid secretion decreases, urates may precipitate in renal tubules and ureters leading to impaction and potentially renal failure.

Renal secondary hyperparathyroidism is common in older animals with a gradual loss of kidney function. As disease progresses, the kidneys are unable to convert calcidiol (25-hydroxycholecalciferol) to the active form, cholecalciferol or vitamin D3, which leads to hypocalcemia and a lack of response to parathyroid hormone, hypertrophy of the parathyroid glands. Decreased glomerular filtration will eventually lead to hyperphosphatemia. Increased phosphorus can lead to the binding of available calcium, decreasing serum calcium concentrations and stimulating production of parathyroid hormone.

Neoplastic

The primary renal tumors reported in reptiles are adenocarcinoma, adenoma, and nephroblastoma. These tumors have been described in lizards, snakes, and chelonians. Grossly one or both kidneys may have irregular masses within the parenchyma. Metastasis is unusual but has been reported.

Nutritional

Nutritional imbalances may include hypovitaminosis A, hypervitaminosis D, hypercalcemia, excess dietary protein, or excess dietary oxalates, as found in spinach or other dark leafy greens. Hypovitaminosis A is most commonly seen in chelonians, and in particular in turtles. Vitamin A deficiency causes the normal epithelium to be replaced by squamous metaplasia and hyperkeratosis. In many reptilian species, this may manifest predominantly as dry eye caused by blockages of the lacrimal glands or aural abscesses in Emydid turtles. However, hypovitaminosis A can also affect other epithelial surfaces, such as the lining of the renal tubules. Squamous metaplasia of the renal tubules can ultimately lead to decreased renal clearance of urates. Hypervitaminosis D has also been associated with metastatic calcification in lizards. High-protein diets rich in purines, most commonly found in dog or cat food, can lead to significant hyperuricemia in herbivorous reptiles that can ultimately result in gout and renal disease. Renal oxalosis has been documented in both health and unhealthy chelonians, and overfeeding plants that contain substantial amounts of oxalates should be avoided.

Degenerative

Glomerulonephrosis and tubulonephrosis, non-specific histopathologic changes characterized as any degenerative non-inflammatory lesions in the kidney, are some of the most common findings in aging reptiles with renal disease.

Gout is a metabolic disease characterized by underexcretion or overproduction of uric acid, resulting in hyperuricemia and deposition of urates in tissues. Any condition that reduces the renal perfusion through dehydration, hemoconcentration, nephrosis, inhibition of glomerular filtration, or renal tubular degeneration many lead to hyperuricemia. In addition, any condition that overloads the animal's ability to process ingested protein, precursor of uric acid, and other nitrogenous metabolic by-products of protein catabolism may predispose an animal to the development of visceral or articular gout. Whereas true gout is caused by the presence of monosodium urate (MSU) crystals, pseudogout occurs as a result of the deposition of crystals other than sodium urate in the joints. Pseudogout is generally described as the presence of calcium crystals in the joints. The most common calcium-containing crystals involved are calcium pyrophosphate dihydrate (CPPD) and basic calcium phosphate crystals that may include carbonated substituted hydroxyapatite, tricalcium phosphate, and octacalcium phosphate. Visceral gout may appear as a white coating when on the capsular surface of affected tissue. Articular gout occurs when uric acid crystals accumulate within the synovial capsules and tendon sheaths of the joints. Clinical signs include lethargy, anorexia, lameness, inability to ambulate, swellings along the joints and organ disfunction.

Urolithiasis (cystic calculi, ureteroliths, cloacal uroliths) has been reported in chelonians and lizards. Cystic calculi are most common. Most uroliths are composed of urate salts, however, calcium-phosphate, -carbonate, -oxalate, and mixed compositions have been reported. It has been postulated that chronic dehydration and high protein diets may be underlying causes.

3. HISTORY AND PHYSICAL EXAMINATION

A complete history and thorough physical examination should be performed. The clinical signs of urinary tract disease are often non-specific to the urinary tract and sometimes include lethargy, weakness, muscle fasciculations, weight loss, coelomic distention and subcutaneous masses or swelling around the joints. In some species, urinary tract disease can lead to constipation or obstipation, dyschezia, bloat, and prolapsed tissue from the vent. Clients infrequently report polyuria and polydipsia. Renomegaly could be palpated via cloaca (e.g. large iguanas) or per cutaneous. In reptiles with articular gout, clinical signs include lethargy, anorexia, lameness, inability to ambulate, swellings along the joints and organ disfunction. Small cystic uroliths in reptiles are unlikely to cause clinical signs. Larger uroliths can result in anorexia, constipation, tenesmus, dystocia, dysuria, poor growth, cloacal prolapse, lethargy, hind limb paresis, and lack of fecal/urine production. Palpation of the prefemoral fossae of chelonians, caudal coelom of lizards, or per cloaca may reveal a hard structure.

4. DIAGNOSTIC TESTS

Complete blood count, plasma biochemistry panel and survey radiography are the minimum data base required when diagnosing urinary disorders. Ancillary diagnostic tests include urinalysis, glomerular filtration rate tests, radiographs, ultrasound, and renal endoscopy and biopsy. Additional tests to detect infectious disease agent (e.g. blood cultures, PCR), as well as toxin screen might be indicated based on the differential diagnoses.

Hematological changes, like normocytic-normochromic anemia, heterophilia, azurophilia, have been reported in reptiles with different type of renal disease but are considered non-specific. Azotemia, hyperuricemia or uremia can be useful as a screening tool to severe renal disease, but the majority of renal function has to be lost for those to be elevated. Uric acid is produced predominantly in the liver but also in smaller amounts in the kidney and pancreas, as the major end product of nitrogen metabolism in most squamates. However, urea and ammonia quantification is important in many chelonians and crocodilians. The relative proportions of ammonia, urea and uric acid may vary with hydration status, postpandrial effects and hepatobiliary disease. Uric acid is cleared mainly via tubular secretion in the kidneys, which is largely independent of glomerular filtration, water resorption and urine flow rate. In severe dehydration or post-renal obstruction, the uric acid can be elevated. Plasma urea in reptiles is produced is excreted by glomerular filtration. In some species, like the green iguana, calcium/phosphorus ratios may become inverted in some cases of renal failure. Plasma electrolyte abnormalities can occur with renal disease in reptiles, but are poorly studied.

Urinalysis rarely performed in reptiles due to extensive post-renal modification of urine by mixing of urine with feces in cloaca and further reabsorption of water in cloaca and rectum. To obtain an uncontaminated, sterile urine sample, the ureters have to be catheterized in the urodeum. This is both technically challenging, and not routinely performed. The urine pH of desert tortoises (*Gopherus agassizii*) is 5.6-7.3, and 6.7 in red-eared sliders. The normal specific gravity is low due to decreased numbers of mammalian nephrons with loops of Henle and lack of counter-current multiplier system. The normal urine is colorless but is easily modified post-renal. Bacteria are normally found (passage through the cloaca) as it is trace of protein and less often glucose. Urate crystals are expected in the sediment, but the presence of cast is always an abnormal finding indicative of tubular disease. Occasionally, protozoan and metazoan parasites can be detected. The presence of blood in urine can be caused by kidney disease, but also could be blood from cloacal, reproductive, or gastrointestinal origins

Glomerular filtration rate (GFR) can be used to quantitate renal function. Plasma exogenous excretion of iohexol has been used following IV administration in the green iguanas and sea turtles to calculate plasma concentrations over 24 hours and the GFR.

Survey radiographs are often use for initial assessment of the urinary tract. Normal reptile kidneys and bladder are often indistinct, especially in chelonians. In lizards, renomegaly can be identified as a soft tissue mass extending cranially from the pelvic canal. Radiopaque uroliths and mineralization might also be identified. Gastrointestinal contrast studies and intravenous urography can help to differentiate the kidneys from surrounding structures. In the case of articular gout, imaging may reveal lytic lesions in or around the joints, often radiolucent if they do not contain calcium.

Ultrasonography of the kidneys and bladder is easily performed in most reptiles. In iguanas, the probe is place cranial or caudal to the pelvic limb with the animal in sternal recumbency, while in bearded dragons the kidneys are best approached dorsally. In chelonians, the approach is through the prefemoral fossa. The kidney parenchyma is uniform because there is not corticomedullar distinction. The thin-walled bladder, in the species with bladder, may contain a large amount of fluid and make differentiation with coelomic effusion difficult. More advance imaging, like CT, would allow evaluation of the urogenital tract and can provide information regarding any additional pathology in the coelomic cavity. Dua-energy computed tomography has been used to characterize the composition of a bladder stone in a desert tortoise before attempting medical management.

Endoscopy and biopsy allow direct visualization of the complete urinary system (kidneys, ureters, bladder in some species and cloaca). Small yellow or white spots within the kidney are signs of urate stasis or gout. A filled and swollen ureter might be due to obstruction or dehydration. Loss of the typical surface structure of the kidney is often a sign of glomerulonephrosis while larger single yellow masses may be nephritis, abscesses or neoplasia. Renal biopsies are necessary to differentiate different conditions. Renal biopsies can be collected endoscopically. Endoscopy can also confirm the diagnosis of cystolith or cloacoliths.

Biopsy of the kidneys in reptiles can also performed via coeliotomy in some species. Surgical access to the chelonian and intrapelvic kidneys of iguanas and bearded dragons can be challenging using standard approaches. In iguanids, a cut-down procedure has been described, through a small longitudinal incision lateral at the tail base. The biopsy sample can be submitted for histopathology and/or bacterial culture and susceptibility.

Fine needle aspirate, cytology and murexide test are needed to confirm the presence of urate tophi, specially in cases of suspected articular gout. Macroscopically, the aspirated material looks like toothpaste and phase contrast microscopy may reveal sharp-needle shape monosuodium urate crystals. The presence of urate can be confirmed by performing the murexide test or by microscopic examination of aspirates from suspected tophi. The murexide test is performed by mixing a drop of nitric acid with a small amount of the suspected material on a slide. The material is dried by evaporation in a flame and allowed to cool. One drop of concentrated ammonia is added, and if urates are present, a pale purple color will develop.

5. TREATMENT

The treatment of the different disorders of the urinary system require addressing the primary cause targeting the therapy to a specific etiology, while providing supportive care to recover from the primary lesions and preventing secondary damage.

Fluid therapy constitutes one of the most important treatments in cases of kidney disorders in reptiles. Uric acid is eliminated by active tubular secretion and water is needed to eliminate the suspension through the renal tubules. Without regular removal by diuresis, urates can accumulate within the kidneys. Fluid type is selected based on results of biochemical analyses, evaluation of blood electrolytes, glucose, and acid–base status. When these values are not known, a balanced isotonic crystalloid solution, such assolution may be used for rehydration and hemodynamic support. Fluids can be given orally, subcutaneously, intracoelomically (IC), IV, or intraosseously. In emergency and critical cases, IV and 10 fluids should be prioritized. Fluids are given at a rate of approximately 1 mL/kg/h to 1.5 mL/kg/h, although that rate can be increased up to 3 mL/kg/h to 5 mL/kg/h for the first 3 hours to 4 hours. For larger reptiles, the placement of an IV catheter or central line is recommended. IC and SC routes may be the only practical options in smaller reptiles, whereas oral fluids should be used for maintenance. SC fluids (20 mL/kg every 12-24 hours) are preferred to IC due to poor absorption in some debilitated animals.

Nutritional support, in the form of syringe feeding or tube feeding may be implemented according to species requirements in order to support the sick or malnourished reptile. The mainstays of dietary management include a reduction in protein (specifically purine) levels and phosphorus along with increasing water intake. Table 1 provides nutritional composition of various food items that may be considered in dietary management of renal disease. To increase water intake, plant materials can be soaked and fed wet, mammal prey can be injected with water, and some herbivores can be fed watery fruits, such as melons 1.

Antibiotics are indicated in cases of confirmed or suspected bacterial nephritis for at least 4-6 weeks. Ideally, antibiotics would be selected based on culture and susceptibility. If this is not possible, broad-spectrum antibiotics are recommended.

Xanthine oxidase inhibitors, such as allopurinol, decrease uric acid synthesis. Allopurinol at 25mg/kg once daily has been shown to decrease uric acid levels by 45% in healthy green iguanas. Uricosuric drugs, such as probenecid and sulfinpyrazone, promote uric acid excretion by the kidneys in mammals mostly by reducing the absorption of uric acid in the tubuli. Reptiles lack the reabsorptive mechanism of uric acid and probenecid could further reduce the active secretion of uric acid into the tubuli so its use is questionable. Uricosuric drugs are also contraindicated when tubular urate crystals are present, which is frequently seen in reptiles.

Phosphate binders, like oral aluminium hydroxide 100 mg/kg PO q12-24 h, could be given to decrease phosphorus if hyperphosphatemia is present, but is recommended to give them between meals so they don’t interfere with calcium absorption. If hypocalcemia is also present, parenteral administration should be avoided to prevent tissue mineralization, unless therapy for tetany or seizures is needed. If so, parenteral calcium should be administered carefully with concomitant diuresis and phosphate binders.

In the case of uroliths and cloacoliths, the treatment includes addressing possible underlying causes (e.g. lack of regular acces to water) and urolith removal via coelitomy and cystotomy or through the cloaca, with or without endoscopic assistance. Medical management with daily soaks, the oral alkalinizing agents potassium citrate and vibration therapy via attachment of a personal massager to the plastron, has been attempted in one desert tortoise with a cystolith and resulted in initial decreased in size of the stone but failed to resolve it long-term.

Table 2
Selected nutritional compositions of various food items that deserve consideration in reptiles with renal disease

Invertebrates	Cricket (*Acheta domesticus*)		Mealworm (*Tenebrio molitor*)	Superworm (*Zophobas morio*)	Silkworm (*Bombyx mori*)	Waxworm (*Galleria mellonella*)	Earthworm (*Lumbricus terrestris*)
	Adult	Juvenile					
Protein (% dry matter)	40–68	40–50	35–55	40–50	65	27–41	73

Vertebrates	Mouse (*Mus musculus*)		Rat (*Rattus norvegicus*)	Meadow Vole (*Microtus pennsylvanicus*)	Smelt (*Sprinchus lanceolatus*)	Herring (*Clupea harengus*)	Chicken (*Gallus gallus domesticus*)	
	Adults	Pups					Adult	Day-old
Protein (% kcal)	48	29	55	63	63	39	47	52

Plants	Romaine Lettuce	Iceberg Lettuce	Alfalfa Sprouts	Mushrooms	Sweet Potato	Squash	Forage & Hays	Prickly Pear	Clover	Apple	Cantaloupe
Protein (% dry matter)	36	25	37	30	5	17	15–17	5	19	1	8

Plants	Lettuce	Cabbage	Cauliflower	Endive	Sweet Potato	Kale	Raisons	Apple	Cantaloupe
Total purine load (mg uric acid/100 g)	13	37	51	17	16	48	107	14	33

Plants	Romaine Lettuce	Iceberg Lettuce	Mustard Greens	Collards	Broccoli	Kale	Spinach
Na (mg/100 g)	8	9	25	18	18	27	79
K (mg/100 g)	290	158	358	169	325	447	558

From Divers SJ and Innis CJ (2019). Urology. In: Divers SJ, Stahl SJ, eds. Maders Reptile and Amphibian Medicine and Surgery, 3rd ed. St. Louis: Elsevier; 2019. pp. 624-648;

REFERENCES [5-10]

6. Divers SJ, Innis CJ. Urology In: Divers SJ,Stahl SJ, eds. *Maders Reptile and Amphibian Medicine and Surgery,*. 3rd ed ed. St. Louis, MO: Elsevier, 2019;624-648.
7. Holz PH. Anatomy and Physiology of the Reptile Renal System. *Vet Clin North Am Exot Anim Pract* 2020;23:103-114.
8. Johnson JG, 3rd, Watson MK. Diseases of the Reptile Renal System. *Vet Clin North Am Exot Anim Pract* 2020;23:115-129.
9. Schmidt L, Di Girolamo N, Selleri P. Diagnostic Imaging of the Reptile Urinary System. *Vet Clin North Am Exot Anim Pract* 2020;23:131-149.
10. Wilkinson SL, Divers SJ. Clinical Management of Reptile Renal Disease. *Vet Clin North Am Exot Anim Pract* 2020;23:151-168.